In [1]:

```python
import numpy as np
import matplotlib.pyplot as plt

import torch
import torch.nn as nn
from torch.utils.data import DataLoader
from torchvision import datasets, transforms

transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Lambda(lambda x: x.view(-1))
])

train_dataset = datasets.FashionMNIST(
    root="./data",
    train=True,
    download=True,
    transform=transform
)

test_dataset = datasets.FashionMNIST(
    root="./data",
    train=False,
    download=True,
    transform=transform
)

train_loader = DataLoader(train_dataset, batch_size=100, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=100, shuffle=False)

class Model1(nn.Module):
    def __init__(self):
        super().__init__()
        self.fc1 = nn.Linear(784, 128)
        self.relu = nn.ReLU()
        self.fc2 = nn.Linear(128, 10)

    def forward(self, x):
        x = self.fc1(x)
        x = self.relu(x)
        x = self.fc2(x)
        return x

class Model2(nn.Module):
    def __init__(self):
        super().__init__()
        self.fc1 = nn.Linear(784, 48)
        self.relu = nn.ReLU()
        self.dropout = nn.Dropout(p=0.2)
        self.fc2 = nn.Linear(48, 10)

    def forward(self, x):
        x = self.fc1(x)
        x = self.relu(x)
        x = self.dropout(x)
        x = self.fc2(x)
        return x
```

```python
def train_one_epoch(model, loader, criterion, optimizer):
    model.train()

    total_loss = 0.0
    correct = 0
    total = 0

    for x, y in loader:
        optimizer.zero_grad()
        outputs = model(x)
        loss = criterion(outputs, y)
        loss.backward()
        optimizer.step()

        total_loss += loss.item() * x.size(0)
        _, predicted = torch.max(outputs, 1)
        correct += (predicted == y).sum().item()
        total += y.size(0)

    avg_loss = total_loss / total
    accuracy = correct / total
    return avg_loss, accuracy

def evaluate(model, loader, criterion):
    model.eval()

    total_loss = 0.0
    correct = 0
    total = 0

    with torch.no_grad():
        for x, y in loader:
            outputs = model(x)
            loss = criterion(outputs, y)

            total_loss += loss.item() * x.size(0)
            _, predicted = torch.max(outputs, 1)
            correct += (predicted == y).sum().item()
            total += y.size(0)

    avg_loss = total_loss / total
    accuracy = correct / total
    return avg_loss, accuracy

num_epochs = 40
criterion = nn.CrossEntropyLoss()

model1 = Model1()
optimizer1 = torch.optim.SGD(model1.parameters(), lr=0.1, weight_decay=0.0)

model1_train_losses = []
model1_test_losses = []
model1_train_accs = []
model1_test_accs = []

for epoch in range(num_epochs):
    train_loss, train_acc = train_one_epoch(model1, train_loader, criterion, opt
    test_loss, test_acc = evaluate(model1, test_loader, criterion)

    model1_train_losses.append(train_loss)
```

```
    model1_test_losses.append(test_loss)
    model1_train_accs.append(train_acc)
    model1_test_accs.append(test_acc)

    print(f"Model 1 - Epoch {epoch+1}/{num_epochs}")
    print(f"Train Loss: {train_loss:.4f}, Train Acc: {train_acc:.4f}")
    print(f"Test  Loss: {test_loss:.4f}, Test  Acc: {test_acc:.4f}")
    print("-" * 40)

model2 = Model2()
optimizer2 = torch.optim.SGD(model2.parameters(), lr=0.1, weight_decay=0.0001)

model2_train_losses = []
model2_test_losses = []
model2_train_accs = []
model2_test_accs = []

for epoch in range(num_epochs):
    train_loss, train_acc = train_one_epoch(model2, train_loader, criterion, opt
    test_loss, test_acc = evaluate(model2, test_loader, criterion)

    model2_train_losses.append(train_loss)
    model2_test_losses.append(test_loss)
    model2_train_accs.append(train_acc)
    model2_test_accs.append(test_acc)

    print(f"Model 2 - Epoch {epoch+1}/{num_epochs}")
    print(f"Train Loss: {train_loss:.4f}, Train Acc: {train_acc:.4f}")
    print(f"Test  Loss: {test_loss:.4f}, Test  Acc: {test_acc:.4f}")
    print("-" * 40)

epochs = range(1, num_epochs + 1)

fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(epochs, model1_train_losses, label="Train Loss")
ax.plot(epochs, model1_test_losses, label="Test Loss")
ax.set_xlabel("Epoch")
ax.set_ylabel("Loss")
ax.set_title("Model 1 Loss")
ax.legend()
ax.grid(True)
plt.show()
plt.close(fig)

fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(epochs, model1_train_accs, label="Train Accuracy")
ax.plot(epochs, model1_test_accs, label="Test Accuracy")
ax.set_xlabel("Epoch")
ax.set_ylabel("Accuracy")
ax.set_title("Model 1 Accuracy")
ax.legend()
ax.grid(True)
plt.show()
plt.close(fig)

fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(epochs, model2_train_losses, label="Train Loss")
ax.plot(epochs, model2_test_losses, label="Test Loss")
ax.set_xlabel("Epoch")
ax.set_ylabel("Loss")
```

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ax.set_title("Model 2 Loss")
ax.legend()
ax.grid(True)
plt.show()
plt.close(fig)

fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(epochs, model2_train_accs, label="Train Accuracy")
ax.plot(epochs, model2_test_accs, label="Test Accuracy")
ax.set_xlabel("Epoch")
ax.set_ylabel("Accuracy")
ax.set_title("Model 2 Accuracy")
ax.legend()
ax.grid(True)
plt.show()
plt.close(fig)

w1_fc1 = model1.fc1.weight.detach().cpu().numpy().flatten()
w1_fc2 = model1.fc2.weight.detach().cpu().numpy().flatten()

fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(w1_fc1, bins=50)
ax.set_title("Model 1 - Input Layer Weights")
ax.set_xlabel("Weight Value")
ax.set_ylabel("Frequency")
plt.show()
plt.close(fig)

fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(w1_fc2, bins=50)
ax.set_title("Model 1 - Hidden/Output Layer Weights")
ax.set_xlabel("Weight Value")
ax.set_ylabel("Frequency")
plt.show()
plt.close(fig)

w2_fc1 = model2.fc1.weight.detach().cpu().numpy().flatten()
w2_fc2 = model2.fc2.weight.detach().cpu().numpy().flatten()

fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(w2_fc1, bins=50)
ax.set_title("Model 2 - Input Layer Weights")
ax.set_xlabel("Weight Value")
ax.set_ylabel("Frequency")
plt.show()
plt.close(fig)

fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(w2_fc2, bins=50)
ax.set_title("Model 2 - Hidden/Output Layer Weights")
ax.set_xlabel("Weight Value")
ax.set_ylabel("Frequency")
plt.show()
plt.close(fig)
```

```
100.0%
100.0%
100.0%
100.0%
```

```
Model 1 - Epoch 1/40
Train Loss: 0.6923, Train Acc: 0.7610
Test  Loss: 0.5229, Test  Acc: 0.8135
----------------------------------------
Model 1 - Epoch 2/40
Train Loss: 0.4690, Train Acc: 0.8348
Test  Loss: 0.4544, Test  Acc: 0.8374
----------------------------------------
Model 1 - Epoch 3/40
Train Loss: 0.4215, Train Acc: 0.8499
Test  Loss: 0.4266, Test  Acc: 0.8510
----------------------------------------
Model 1 - Epoch 4/40
Train Loss: 0.3940, Train Acc: 0.8580
Test  Loss: 0.4286, Test  Acc: 0.8458
----------------------------------------
Model 1 - Epoch 5/40
Train Loss: 0.3722, Train Acc: 0.8668
Test  Loss: 0.4219, Test  Acc: 0.8498
----------------------------------------
Model 1 - Epoch 6/40
Train Loss: 0.3563, Train Acc: 0.8719
Test  Loss: 0.3954, Test  Acc: 0.8618
----------------------------------------
Model 1 - Epoch 7/40
Train Loss: 0.3448, Train Acc: 0.8757
Test  Loss: 0.3991, Test  Acc: 0.8607
----------------------------------------
Model 1 - Epoch 8/40
Train Loss: 0.3329, Train Acc: 0.8806
Test  Loss: 0.3771, Test  Acc: 0.8656
----------------------------------------
Model 1 - Epoch 9/40
Train Loss: 0.3232, Train Acc: 0.8849
Test  Loss: 0.3703, Test  Acc: 0.8678
----------------------------------------
Model 1 - Epoch 10/40
Train Loss: 0.3151, Train Acc: 0.8860
Test  Loss: 0.3675, Test  Acc: 0.8720
----------------------------------------
Model 1 - Epoch 11/40
Train Loss: 0.3085, Train Acc: 0.8878
Test  Loss: 0.3536, Test  Acc: 0.8708
----------------------------------------
Model 1 - Epoch 12/40
Train Loss: 0.2998, Train Acc: 0.8919
Test  Loss: 0.3881, Test  Acc: 0.8622
----------------------------------------
Model 1 - Epoch 13/40
Train Loss: 0.2938, Train Acc: 0.8930
Test  Loss: 0.3549, Test  Acc: 0.8734
----------------------------------------
Model 1 - Epoch 14/40
Train Loss: 0.2881, Train Acc: 0.8953
Test  Loss: 0.3445, Test  Acc: 0.8751
----------------------------------------
Model 1 - Epoch 15/40
Train Loss: 0.2826, Train Acc: 0.8978
Test  Loss: 0.3520, Test  Acc: 0.8724
----------------------------------------
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```
Model 1 - Epoch 16/40
Train Loss: 0.2779, Train Acc: 0.8980
Test  Loss: 0.3516, Test  Acc: 0.8736
----------------------------------------
Model 1 - Epoch 17/40
Train Loss: 0.2714, Train Acc: 0.9013
Test  Loss: 0.3509, Test  Acc: 0.8743
----------------------------------------
Model 1 - Epoch 18/40
Train Loss: 0.2671, Train Acc: 0.9028
Test  Loss: 0.3421, Test  Acc: 0.8750
----------------------------------------
Model 1 - Epoch 19/40
Train Loss: 0.2617, Train Acc: 0.9043
Test  Loss: 0.3428, Test  Acc: 0.8764
----------------------------------------
Model 1 - Epoch 20/40
Train Loss: 0.2580, Train Acc: 0.9068
Test  Loss: 0.3361, Test  Acc: 0.8822
----------------------------------------
Model 1 - Epoch 21/40
Train Loss: 0.2536, Train Acc: 0.9074
Test  Loss: 0.3465, Test  Acc: 0.8737
----------------------------------------
Model 1 - Epoch 22/40
Train Loss: 0.2506, Train Acc: 0.9087
Test  Loss: 0.3352, Test  Acc: 0.8798
----------------------------------------
Model 1 - Epoch 23/40
Train Loss: 0.2458, Train Acc: 0.9101
Test  Loss: 0.3312, Test  Acc: 0.8844
----------------------------------------
Model 1 - Epoch 24/40
Train Loss: 0.2425, Train Acc: 0.9113
Test  Loss: 0.3313, Test  Acc: 0.8819
----------------------------------------
Model 1 - Epoch 25/40
Train Loss: 0.2386, Train Acc: 0.9133
Test  Loss: 0.3232, Test  Acc: 0.8858
----------------------------------------
Model 1 - Epoch 26/40
Train Loss: 0.2361, Train Acc: 0.9139
Test  Loss: 0.3551, Test  Acc: 0.8686
----------------------------------------
Model 1 - Epoch 27/40
Train Loss: 0.2331, Train Acc: 0.9150
Test  Loss: 0.3362, Test  Acc: 0.8796
----------------------------------------
Model 1 - Epoch 28/40
Train Loss: 0.2290, Train Acc: 0.9165
Test  Loss: 0.3388, Test  Acc: 0.8807
----------------------------------------
Model 1 - Epoch 29/40
Train Loss: 0.2255, Train Acc: 0.9170
Test  Loss: 0.3226, Test  Acc: 0.8886
----------------------------------------
Model 1 - Epoch 30/40
Train Loss: 0.2216, Train Acc: 0.9184
Test  Loss: 0.3276, Test  Acc: 0.8850
----------------------------------------
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Model 1 - Epoch 31/40
Train Loss: 0.2199, Train Acc: 0.9196
Test  Loss: 0.3312, Test  Acc: 0.8860
----------------------------------------
Model 1 - Epoch 32/40
Train Loss: 0.2153, Train Acc: 0.9225
Test  Loss: 0.3340, Test  Acc: 0.8841
----------------------------------------
Model 1 - Epoch 33/40
Train Loss: 0.2136, Train Acc: 0.9236
Test  Loss: 0.3342, Test  Acc: 0.8853
----------------------------------------
Model 1 - Epoch 34/40
Train Loss: 0.2113, Train Acc: 0.9230
Test  Loss: 0.3295, Test  Acc: 0.8844
----------------------------------------
Model 1 - Epoch 35/40
Train Loss: 0.2083, Train Acc: 0.9239
Test  Loss: 0.3219, Test  Acc: 0.8865
----------------------------------------
Model 1 - Epoch 36/40
Train Loss: 0.2056, Train Acc: 0.9254
Test  Loss: 0.3268, Test  Acc: 0.8880
----------------------------------------
Model 1 - Epoch 37/40
Train Loss: 0.2036, Train Acc: 0.9269
Test  Loss: 0.3367, Test  Acc: 0.8832
----------------------------------------
Model 1 - Epoch 38/40
Train Loss: 0.2005, Train Acc: 0.9276
Test  Loss: 0.3203, Test  Acc: 0.8909
----------------------------------------
Model 1 - Epoch 39/40
Train Loss: 0.1987, Train Acc: 0.9277
Test  Loss: 0.3251, Test  Acc: 0.8880
----------------------------------------
Model 1 - Epoch 40/40
Train Loss: 0.1944, Train Acc: 0.9295
Test  Loss: 0.3288, Test  Acc: 0.8866
----------------------------------------
Model 2 - Epoch 1/40
Train Loss: 0.7594, Train Acc: 0.7391
Test  Loss: 0.5466, Test  Acc: 0.8066
----------------------------------------
Model 2 - Epoch 2/40
Train Loss: 0.5193, Train Acc: 0.8168
Test  Loss: 0.4770, Test  Acc: 0.8282
----------------------------------------
Model 2 - Epoch 3/40
Train Loss: 0.4706, Train Acc: 0.8343
Test  Loss: 0.4394, Test  Acc: 0.8407
----------------------------------------
Model 2 - Epoch 4/40
Train Loss: 0.4439, Train Acc: 0.8424
Test  Loss: 0.4461, Test  Acc: 0.8343
----------------------------------------
Model 2 - Epoch 5/40
Train Loss: 0.4255, Train Acc: 0.8483
Test  Loss: 0.4208, Test  Acc: 0.8482
----------------------------------------
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Model 2 - Epoch 6/40
Train Loss: 0.4112, Train Acc: 0.8521
Test  Loss: 0.4022, Test  Acc: 0.8524
----------------------------------------
Model 2 - Epoch 7/40
Train Loss: 0.3998, Train Acc: 0.8584
Test  Loss: 0.3954, Test  Acc: 0.8559
----------------------------------------
Model 2 - Epoch 8/40
Train Loss: 0.3913, Train Acc: 0.8609
Test  Loss: 0.3936, Test  Acc: 0.8598
----------------------------------------
Model 2 - Epoch 9/40
Train Loss: 0.3821, Train Acc: 0.8633
Test  Loss: 0.3840, Test  Acc: 0.8594
----------------------------------------
Model 2 - Epoch 10/40
Train Loss: 0.3784, Train Acc: 0.8630
Test  Loss: 0.3808, Test  Acc: 0.8624
----------------------------------------
Model 2 - Epoch 11/40
Train Loss: 0.3721, Train Acc: 0.8656
Test  Loss: 0.3804, Test  Acc: 0.8647
----------------------------------------
Model 2 - Epoch 12/40
Train Loss: 0.3668, Train Acc: 0.8678
Test  Loss: 0.3888, Test  Acc: 0.8577
----------------------------------------
Model 2 - Epoch 13/40
Train Loss: 0.3631, Train Acc: 0.8682
Test  Loss: 0.3736, Test  Acc: 0.8661
----------------------------------------
Model 2 - Epoch 14/40
Train Loss: 0.3582, Train Acc: 0.8716
Test  Loss: 0.3833, Test  Acc: 0.8596
----------------------------------------
Model 2 - Epoch 15/40
Train Loss: 0.3528, Train Acc: 0.8726
Test  Loss: 0.3685, Test  Acc: 0.8669
----------------------------------------
Model 2 - Epoch 16/40
Train Loss: 0.3501, Train Acc: 0.8734
Test  Loss: 0.3701, Test  Acc: 0.8649
----------------------------------------
Model 2 - Epoch 17/40
Train Loss: 0.3467, Train Acc: 0.8743
Test  Loss: 0.3687, Test  Acc: 0.8667
----------------------------------------
Model 2 - Epoch 18/40
Train Loss: 0.3423, Train Acc: 0.8755
Test  Loss: 0.3671, Test  Acc: 0.8679
----------------------------------------
Model 2 - Epoch 19/40
Train Loss: 0.3418, Train Acc: 0.8770
Test  Loss: 0.3608, Test  Acc: 0.8707
----------------------------------------
Model 2 - Epoch 20/40
Train Loss: 0.3393, Train Acc: 0.8769
Test  Loss: 0.3597, Test  Acc: 0.8704
----------------------------------------
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```
Model 2 - Epoch 21/40
Train Loss: 0.3352, Train Acc: 0.8783
Test  Loss: 0.3595, Test  Acc: 0.8705
----------------------------------------
Model 2 - Epoch 22/40
Train Loss: 0.3334, Train Acc: 0.8785
Test  Loss: 0.3676, Test  Acc: 0.8697
----------------------------------------
Model 2 - Epoch 23/40
Train Loss: 0.3297, Train Acc: 0.8799
Test  Loss: 0.3520, Test  Acc: 0.8745
----------------------------------------
Model 2 - Epoch 24/40
Train Loss: 0.3287, Train Acc: 0.8793
Test  Loss: 0.3605, Test  Acc: 0.8680
----------------------------------------
Model 2 - Epoch 25/40
Train Loss: 0.3263, Train Acc: 0.8812
Test  Loss: 0.3559, Test  Acc: 0.8733
----------------------------------------
Model 2 - Epoch 26/40
Train Loss: 0.3226, Train Acc: 0.8813
Test  Loss: 0.3561, Test  Acc: 0.8716
----------------------------------------
Model 2 - Epoch 27/40
Train Loss: 0.3205, Train Acc: 0.8833
Test  Loss: 0.3534, Test  Acc: 0.8754
----------------------------------------
Model 2 - Epoch 28/40
Train Loss: 0.3213, Train Acc: 0.8837
Test  Loss: 0.3710, Test  Acc: 0.8653
----------------------------------------
Model 2 - Epoch 29/40
Train Loss: 0.3189, Train Acc: 0.8834
Test  Loss: 0.3464, Test  Acc: 0.8765
----------------------------------------
Model 2 - Epoch 30/40
Train Loss: 0.3179, Train Acc: 0.8838
Test  Loss: 0.3500, Test  Acc: 0.8747
----------------------------------------
Model 2 - Epoch 31/40
Train Loss: 0.3151, Train Acc: 0.8849
Test  Loss: 0.3520, Test  Acc: 0.8746
----------------------------------------
Model 2 - Epoch 32/40
Train Loss: 0.3143, Train Acc: 0.8845
Test  Loss: 0.3496, Test  Acc: 0.8762
----------------------------------------
Model 2 - Epoch 33/40
Train Loss: 0.3121, Train Acc: 0.8849
Test  Loss: 0.3501, Test  Acc: 0.8741
----------------------------------------
Model 2 - Epoch 34/40
Train Loss: 0.3117, Train Acc: 0.8863
Test  Loss: 0.3507, Test  Acc: 0.8723
----------------------------------------
Model 2 - Epoch 35/40
Train Loss: 0.3073, Train Acc: 0.8872
Test  Loss: 0.3430, Test  Acc: 0.8784
----------------------------------------
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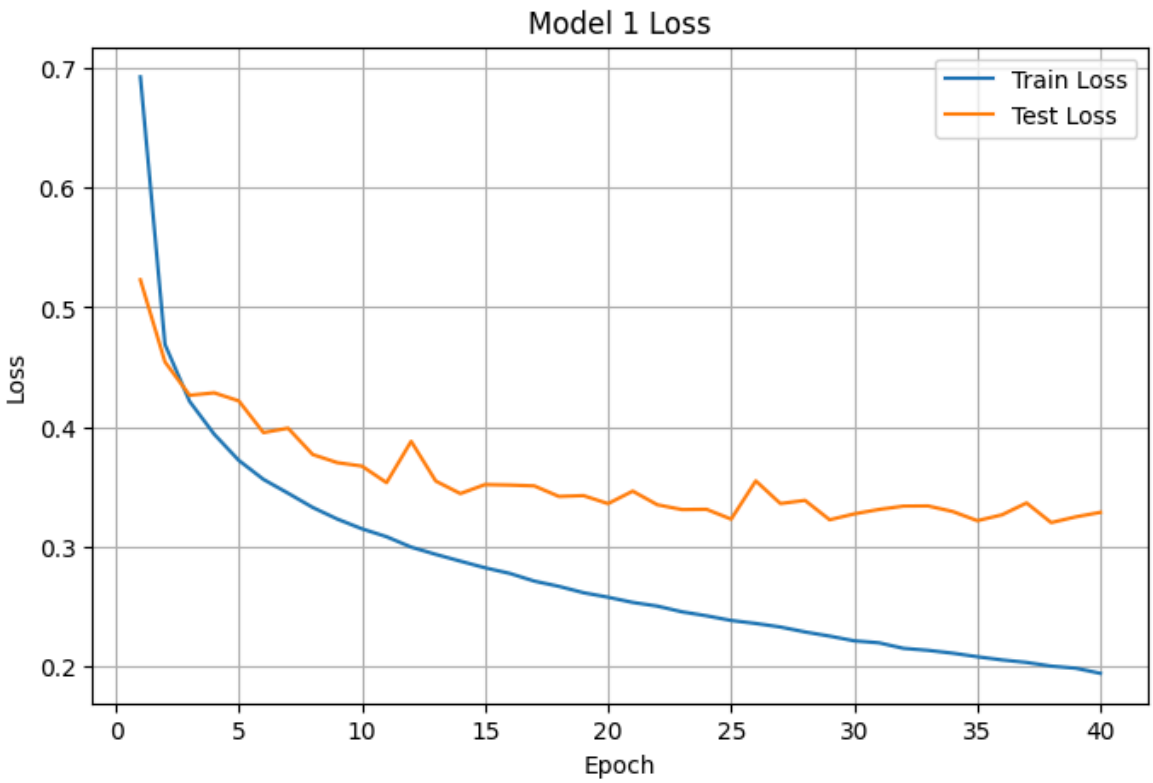
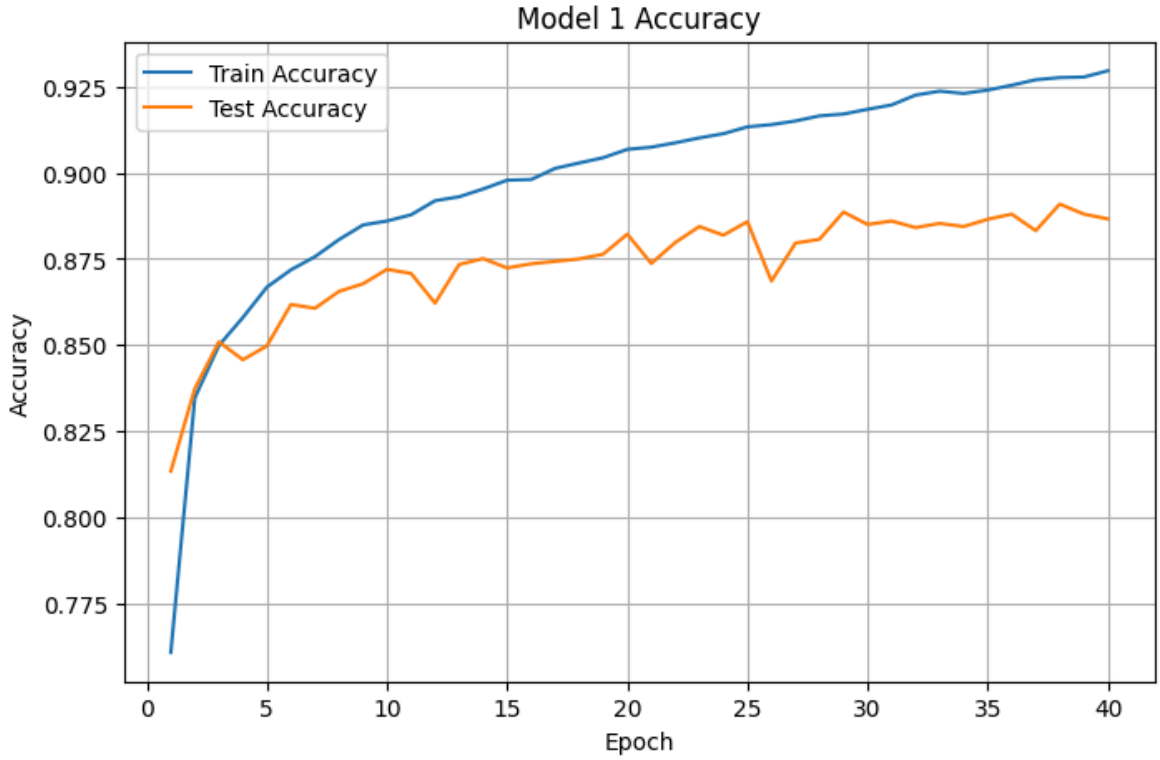
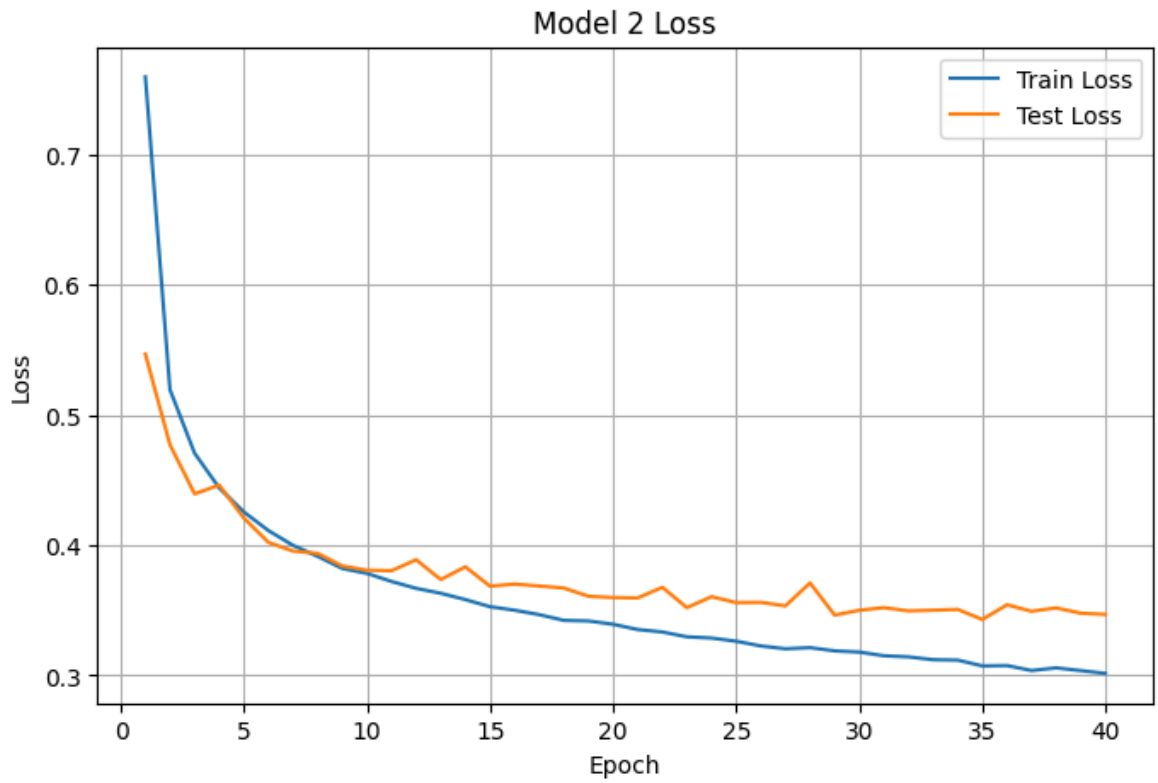
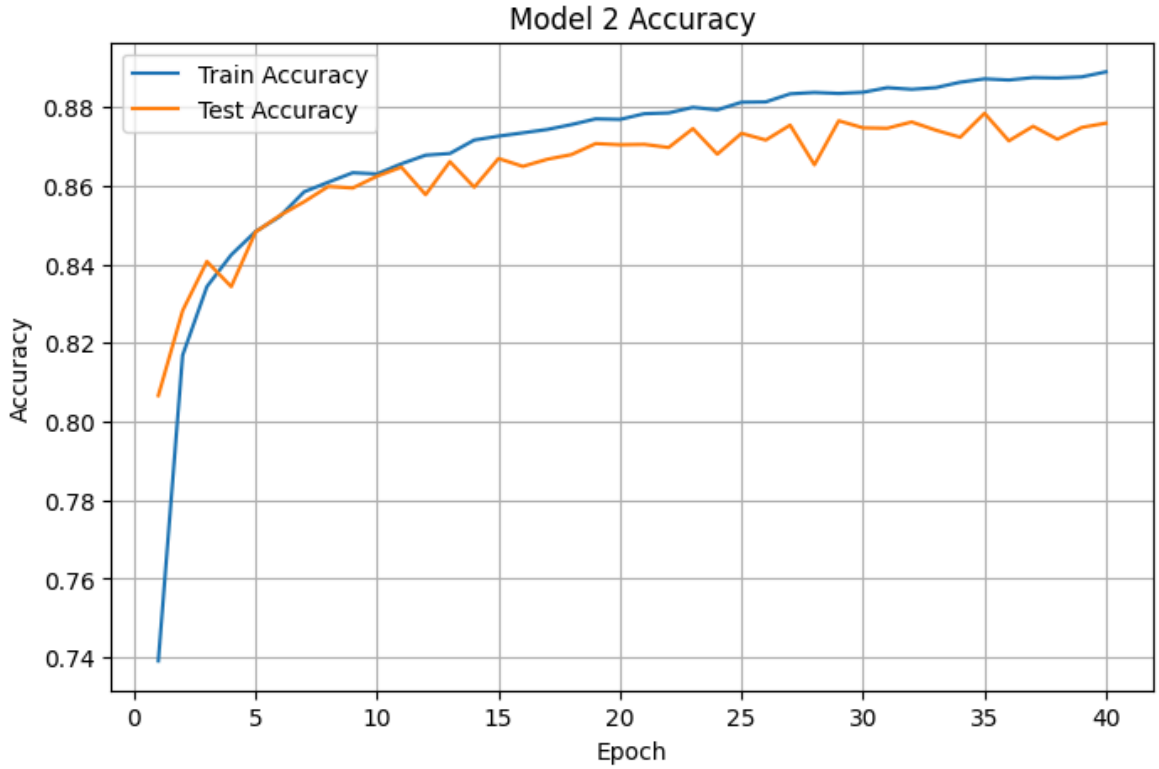
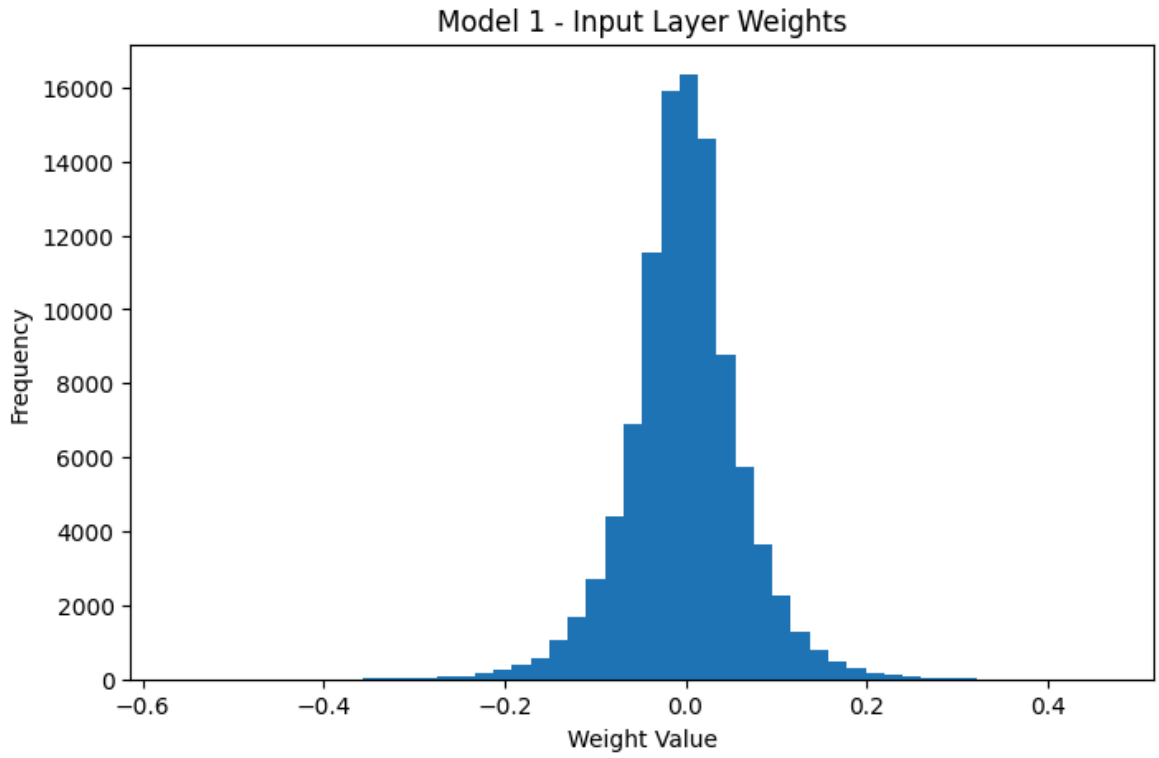
Model 2 - Epoch 36/40
Train Loss: 0.3076, Train Acc: 0.8869
Test  Loss: 0.3542, Test  Acc: 0.8714
----------------------------------------
Model 2 - Epoch 37/40
Train Loss: 0.3038, Train Acc: 0.8875
Test  Loss: 0.3493, Test  Acc: 0.8751
----------------------------------------
Model 2 - Epoch 38/40
Train Loss: 0.3058, Train Acc: 0.8874
Test  Loss: 0.3518, Test  Acc: 0.8718
----------------------------------------
Model 2 - Epoch 39/40
Train Loss: 0.3037, Train Acc: 0.8877
Test  Loss: 0.3478, Test  Acc: 0.8748
----------------------------------------
Model 2 - Epoch 40/40
Train Loss: 0.3016, Train Acc: 0.8890
Test  Loss: 0.3469, Test  Acc: 0.8759
----------------------------------------
```

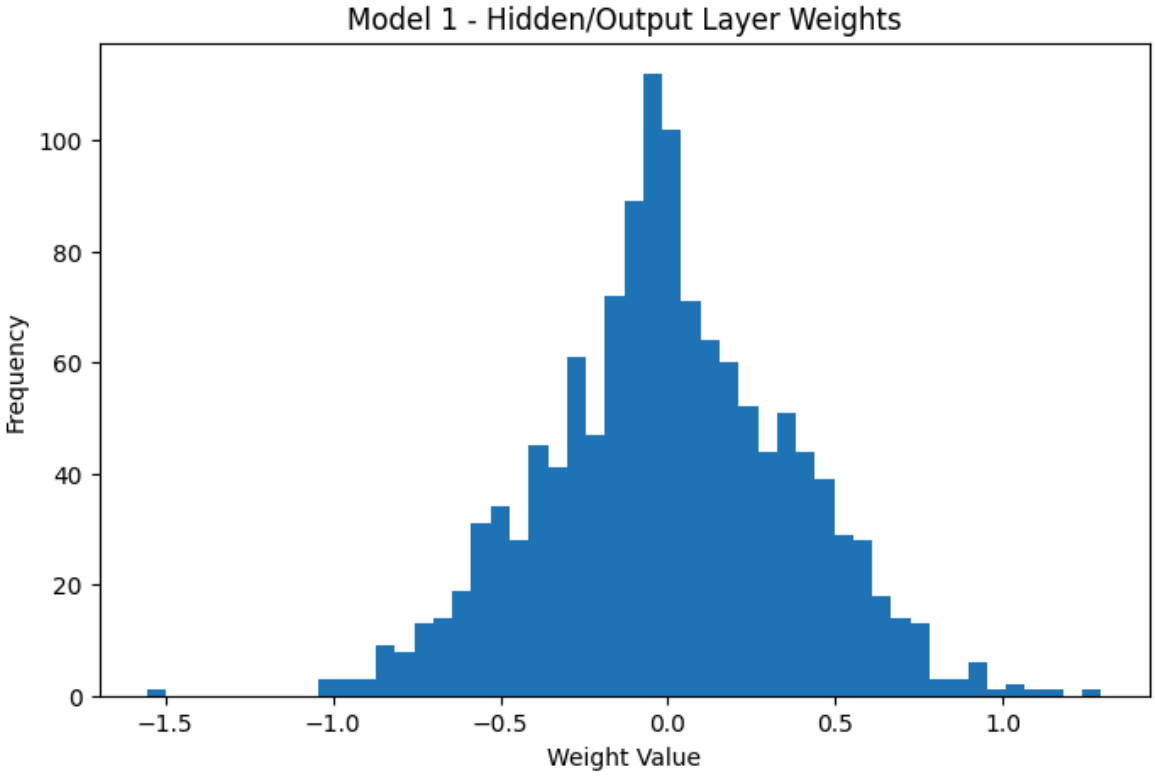


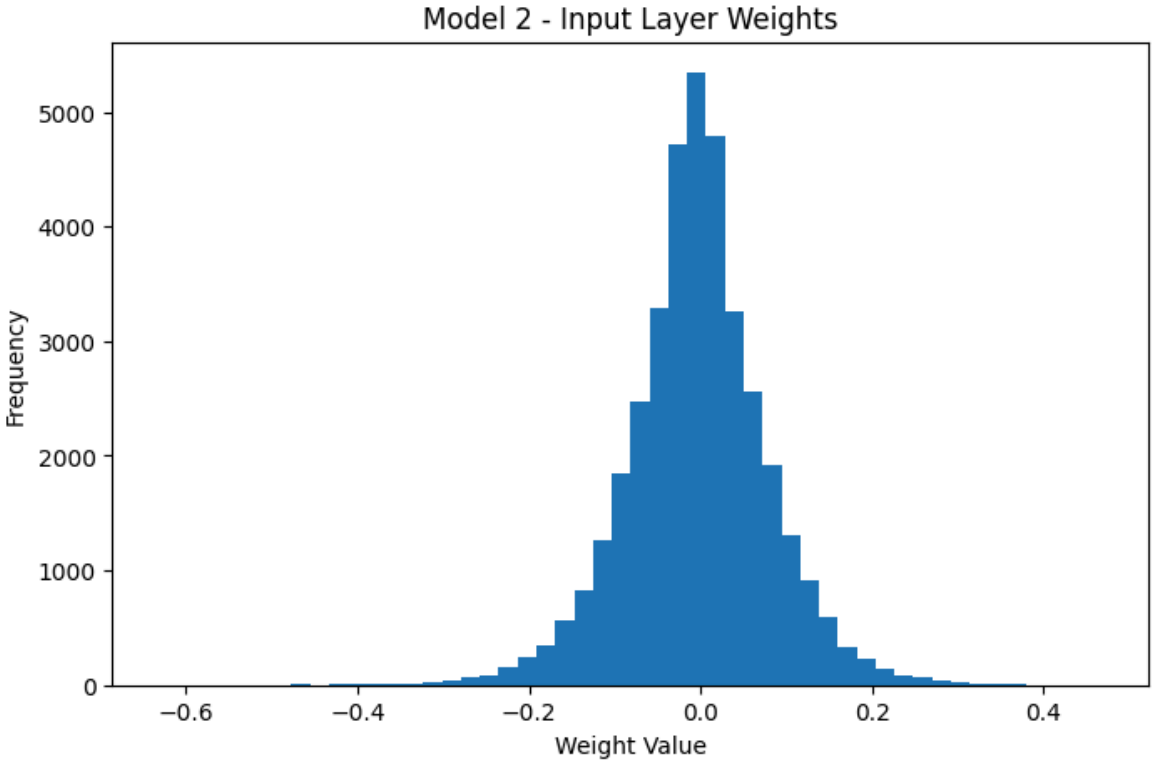

Model 1 Accuracy
Train Accuracy
Test Accuracy
Accuracy
Epoch
0.925
0.900
0.875
0.850
0.825
0.800
0.775
0
5
10
15
20
25
30
35
40

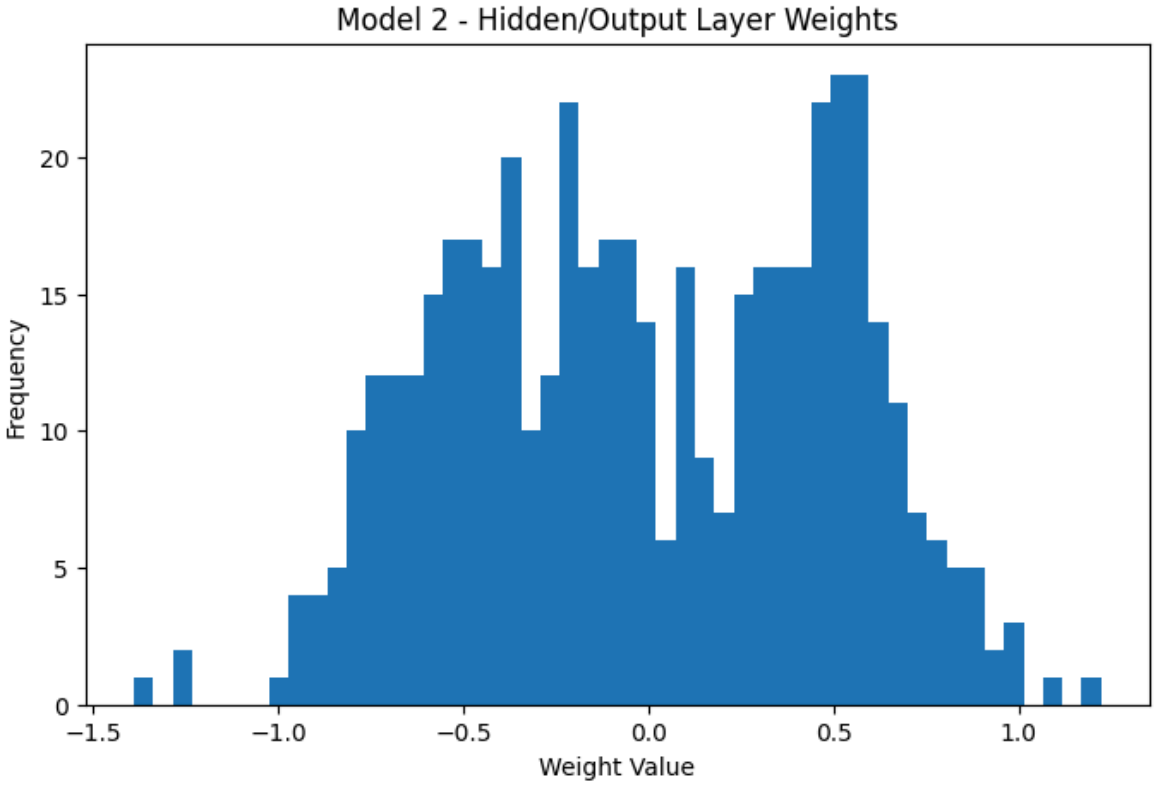

Model 2 Loss
Train Loss
Test Loss
Loss
Epoch
0.7
0.6
0.5
0.4
0.3
0
5
10
15
20
25
30
35
40


Model 2 Accuracy
Train Accuracy
Test Accuracy
Accuracy
Epoch
0.88
0.86
0.84
0.82
0.80
0.78
0.76
0.74
0
5
10
15
20
25
30
35
40


Model 1 - Input Layer Weights
Frequency
Weight Value
16000
14000
12000
10000
8000
6000
4000
2000
0
−0.6
−0.4
−0.2
0.0
0.2
0.4


Model 1 - Hidden/Output Layer Weights
Frequency
Weight Value
0
20
40
60
80
100
−1.5
−1.0
−0.5
0.0
0.5
1.0


Model 2 - Input Layer Weights
Frequency
Weight Value
0
1000
2000
3000
4000
5000
−0.6
−0.4
−0.2
0.0
0.2
0.4



Model comparison.

Two neural networks were trained on Fashion MNIST for 40 epochs. Model 1 used one hidden layer with 128 ReLU units and did not use regularization or dropout. Model 2 used one hidden layer with 48 ReLU units, L2 regularization with coefficient 0.0001, and dropout with rate 0.2. Both models were trained using SGD and cross-entropy loss.

Training result analysis

Model 1 achieved higher training accuracy and lower training loss than Model 2. However, the gap between training and test performance was also larger, which suggests that Model 1 had a stronger tendency to overfit. In contrast, Model 2 had slightly lower training performance, but the training and test curves stayed closer together. This indicates that regularization and dropout helped control overfitting and improved generalization stability.

Histogram comparison

The weight histograms of Model 1 are more spread out, with a wider range of positive and negative values. In particular, the hidden/output layer of Model 1 contains more large-magnitude weights. In contrast, the weights of Model 2 are more concentrated around zero, especially in the input layer histogram. The distribution is narrower and contains fewer extreme values.

Effect of regularization

Regularization makes the weight distribution more concentrated around zero. It reduces the magnitude of the weights and discourages excessively large positive

or negative values. As a result, the model becomes less complex and is less likely to overfit the training data. Therefore, regularization helps improve generalization by shrinking the weights toward zero.

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